

BCA III SEM – STRINGS (C#)

2 MARKS QUESTIONS

1. Differentiate immutable and mutable strings

Immutable strings cannot be changed once created (String).
Mutable strings can be modified after creation (StringBuilder).

2. What are verbatim strings?

Verbatim strings use @ symbol and ignore escape characters.
Example: @"C:\Folder\File"

3. What is the use of ToString() method?

Converts an object or value into string representation.

4. What are mutable strings? How are they created?

Mutable strings allow modification of content.
Created using StringBuilder class.

5. Insert() method

Used to insert a string at a specified position.
Syntax: string.Insert(index, value)

6. Substring() method

Used to extract a part of a string.
Syntax: string.Substring(startIndex, length)

7. Array of strings

Used to store multiple string values.
Example: string[] names = { "A", "B", "C" };

5 MARKS QUESTIONS

1. Explain the different ways of creating strings in C# with example

1) Using string literal

Example: `string s1 = "Hello";`

2) Using string constructor

Example: `string s2 = new string('A', 3);`

3) Using string concatenation

Example: `string s3 = "Hello" + " World";`

4) Using `String.Format()` method

Example: `string s4 = String.Format("Age is {0}", 20);`

5) Using verbatim string (`@`)

Example: `string path = @"C:\Folder\File";`

2. Explain any five methods of `String` class

1) `Length` – Returns number of characters in a string.

2) `ToUpper()` – Converts string to uppercase.

3) `ToLower()` – Converts string to lowercase.

4) `Substring()` – Extracts a part of string.

5) `Replace()` – Replaces characters or substring.

3. Explain the methods of `StringBuilder` class

1) `Append()` – Adds text at the end.

2) `Insert()` – Inserts text at specified position.

3) `Remove()` – Removes characters from string.

4) `Replace()` – Replaces characters or substring.

5) `Clear()` – Removes all characters.